

SentinelNet - AI-powered Intelligent Intrusion Detection Using Machine Learning

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ABSTRACT

With the rapid expansion of digital infrastructures, safeguarding network environments has become a pressing challenge. Traditional defense mechanisms such as firewalls and signature-based antivirus tools often fail to recognize advanced, evolving threats. This project, NetSecurify, explores the use of machine learning for building an adaptive Intrusion Detection System (IDS) that can detect both known and novel attacks in real time. Using two benchmark datasets—NSL-KDD (representing legacy attacks) and CICIDS2017 (representing modern traffic patterns)—we develop and test models ranging from Logistic Regression to ensemble methods like Random Forest and Gradient Boosting. The preprocessing workflow focuses on handling class imbalance, feature scaling, and meaningful feature selection to improve detection capability. Our experiments demonstrate that ensemble approaches achieve detection rates above 99% on CICIDS2017 with efficient computation, highlighting their viability for enterprise-level IDS deployment. This work contributes to cybersecurity by presenting a comparative analysis of multiple algorithms and emphasizing the balance between detection accuracy and system performance.

Keywords: Intrusion Detection, Cybersecurity, Machine Learning, NSL-KDD, CICIDS2017, Random Forest, Gradient Boosting

1. INTRODUCTION

The digitization of critical services and the proliferation of internet-connected devices have drastically reshaped the threat landscape. Cyber adversaries today employ sophisticated strategies ranging from Distributed Denial of Service (DDoS) attacks to stealthy Advanced Persistent Threats (APTs). In this dynamic environment, static security measures prove insufficient. There is a need for intelligent systems capable of recognizing anomalous behaviors and adapting to previously unseen attack patterns.

Intrusion Detection Systems (IDS) are a cornerstone of layered defense mechanisms, continuously analyzing network traffic for suspicious activity. Unlike preventive controls, IDS provides visibility into live attacks, enabling faster detection and

mitigation. Incorporating machine learning into IDS design allows the system to learn complex traffic patterns and generalize across different attack categories.

The NetSecurify project aims to design such an IDS by combining signature-based and anomaly-based detection principles with machine learning algorithms. Core objectives include:

- Automated Threat Classification
- Real-Time Detection
- Multi-Attack Recognition
- Handling Class Imbalance
- Benchmarking Algorithms

2. LITERATURE REVIEW

Early IDS solutions were primarily signature-driven, detecting threats based on predefined patterns. While efficient for known malware, such systems could not cope with zero-day exploits or polymorphic malware. This gap led to anomaly-based approaches, which use statistical learning and machine learning to model “normal” behavior and flag deviations.

Machine learning has significantly influenced IDS evolution. Decision Trees became popular for their interpretability, while Support Vector Machines offered robustness in binary classification tasks. Later, ensemble methods like Random Forest and Gradient Boosting achieved state-of-the-art results by combining multiple learners.

Datasets have played a key role in IDS research. The NSL-KDD dataset remains a widely used benchmark, though its simulated attacks are outdated. The CICIDS2017 dataset, by contrast, offers realistic enterprise traffic including modern threats like web application exploits, botnet communication, and infiltration attempts. Together, these datasets enable comprehensive testing of IDS performance across legacy and modern contexts.

Recent trends point toward hybrid IDS architectures, where signature detection provides quick filtering while anomaly detection captures emerging threats. This project aligns with these trends by implementing a hybrid, ML-powered IDS.

3. METHODOLOGY

The development of NetSecurify followed a structured methodology:

- Datasets Used:
 - NSL-KDD: 41 features across ~150,000 records, including DoS, Probe, U2R, and R2L attack categories.
 - CICIDS2017: Over 2.8 million records with 80+ features, covering modern attack types such as DDoS variants, brute force attempts, SQL injection, and botnet activity.
- Preprocessing:
 - Removal of duplicates and handling of missing values.
 - Standardization (Z-score scaling) applied to continuous features.
 - One-hot encoding for categorical attributes like protocol type and service.
 - Feature selection using correlation analysis and feature importance rankings.
 - Addressing class imbalance through undersampling and power sampling, avoiding synthetic data generation like SMOTE.
- Models Evaluated:
 - Logistic Regression (baseline).
 - Decision Tree.
 - Random Forest.
 - Gradient Boosting.

Performance was assessed on binary (normal vs. attack) and multi-class (specific attack types) tasks using metrics such as accuracy, precision, recall, F1-score, specificity, and ROC-AUC.

4. RESULTS

- Binary Classification:
 - On NSL-KDD, Random Forest and Gradient Boosting outperformed Logistic Regression, achieving testing accuracies around 86%.
 - On CICIDS2017, ensemble methods reached nearly perfect classification (~99.8% accuracy).
- Multi-Class Classification:
 - NSL-KDD showed the challenges of multi-class classification, with ensemble models achieving around 86% accuracy.

- CICIDS2017 demonstrated the strength of Gradient Boosting, reaching above 99% accuracy.

5. CONCLUSION

The NetSecurify framework highlights the advantages of ensemble machine learning models in intrusion detection. Results indicate that while Logistic Regression is suitable for lightweight evaluation, Random Forest and Gradient Boosting are better suited for realistic deployments, balancing accuracy and computational demands.

Binary classification offers quick detection with minimal overhead, while multi-class classification provides valuable granularity at higher computational cost. The CICIDS2017 dataset proved more effective than NSL-KDD for evaluating models against modern threats.

Future extensions could explore deep learning architectures (CNNs, LSTMs) and real-time IDS deployment integrated with cloud monitoring systems for scalable enterprise defense.

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